

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	27/07/2026	Duration:	60 minutes

Code of Conduct

I Dhanushree S-192421218 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Dhanushree S

Annexure A – Agile Sprint Planning & User Story Workshop

AI-Based Crop Disease Detection and Farmer Advisory Platform

1.1 Introduction

Agriculture is one of the most significant sectors supporting the economy of many developing countries. It provides employment to millions of people, contributes to food security, and plays an essential role in economic development. Farmers depend on healthy crops for obtaining maximum agricultural productivity and profit. However, crop diseases caused by fungi, bacteria, viruses, pests, nutrient deficiencies, and environmental conditions are major threats to agricultural production. Every year, farmers suffer heavy financial losses because diseases are identified at later stages when crops are already severely damaged.

Traditionally, crop disease identification depends on experienced farmers or agricultural experts who inspect crops manually. This process requires considerable time, expertise, and financial resources. Farmers living in remote villages often do not have immediate access to agricultural specialists. Consequently, diseases spread rapidly before proper treatment can be applied. Incorrect diagnosis frequently results in excessive pesticide usage, increased farming expenses, environmental pollution, soil degradation, and lower crop quality.

Advancements in Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Computer Vision have transformed agriculture by introducing intelligent crop monitoring systems. AI-based disease detection systems analyze crop leaf images using trained deep learning models and accurately identify diseases within seconds. Farmers can simply capture images of crop leaves using smartphones and receive instant disease diagnosis and treatment recommendations without waiting for expert consultation.

The AI-Based Crop Disease Detection and Farmer Advisory Platform is proposed to address these agricultural challenges through an intelligent software solution. The platform enables farmers to upload crop images, automatically detect diseases, estimate prediction confidence, and receive personalized recommendations regarding pesticides, fertilizers, irrigation schedules, preventive measures, and crop maintenance. The platform also maintains historical disease records, generates agricultural reports, and supports communication between farmers and agricultural experts.

The software follows the Agile Software Development Methodology, allowing iterative development through multiple sprints. Continuous stakeholder feedback, sprint planning, backlog prioritization, sprint reviews, and retrospective meetings ensure continuous improvement of the software product. Agile methodology allows developers to quickly adapt to changing farmer requirements while maintaining software quality and customer satisfaction.

The proposed system contributes to smart agriculture by combining Artificial Intelligence, cloud computing, mobile applications, agricultural knowledge, and expert advisory services into a single integrated platform. It helps farmers reduce crop losses, improve agricultural productivity, decrease unnecessary pesticide usage, and promote sustainable farming practices.

1.2 Project Vision

The vision of the AI-Based Crop Disease Detection and Farmer Advisory Platform is to create a smart, intelligent, and user-friendly digital agriculture ecosystem that empowers farmers through advanced Artificial Intelligence technologies. The system aims to make accurate crop disease diagnosis easily accessible to farmers regardless of their educational background, technical knowledge, or geographical location.

The platform envisions replacing traditional manual disease diagnosis with intelligent AI-powered image analysis capable of detecting crop diseases during their early stages. Early diagnosis enables farmers to take immediate corrective actions, reducing disease spread and minimizing financial losses.

Another important vision is to build an integrated agricultural advisory platform that not only detects diseases but also provides scientifically validated recommendations regarding pesticide selection, fertilizer application, irrigation scheduling, nutrient management, soil improvement, pest control, and preventive farming practices. Farmers receive personalized recommendations based on crop type, disease severity, environmental conditions, and agricultural best practices.

The long-term vision extends beyond disease detection by integrating weather forecasting, IoT sensors, satellite imagery, drone surveillance, blockchain-based agricultural records, multilingual virtual assistants, and predictive analytics. Such integration will transform traditional farming into intelligent precision agriculture capable of improving productivity while protecting natural resources.

The platform also supports government agricultural departments by providing disease monitoring dashboards, outbreak prediction reports, regional crop statistics, and analytical insights that assist in policymaking and agricultural planning. Researchers can utilize anonymous disease datasets for developing improved machine learning models and conducting agricultural research.

Ultimately, the project seeks to improve farmers' income, enhance food security, reduce environmental pollution, encourage sustainable agriculture, and contribute to national economic development through intelligent farming technologies.

1.3 Project Objectives

The primary objective of the project is to design and develop an Artificial Intelligence-enabled software platform capable of automatically detecting crop diseases from leaf images and providing comprehensive advisory services to farmers.

The specific objectives are as follows:

Objective 1: Early Disease Detection

Develop an intelligent disease detection system capable of identifying crop diseases during their initial stages, thereby preventing widespread crop damage.

Objective 2: AI-Based Image Classification

Design and train deep learning models that accurately classify healthy crops and multiple disease categories using leaf images.

Objective 3: Farmer-Friendly Mobile Application

Develop an easy-to-use mobile and web application allowing farmers to upload crop images without requiring technical expertise.

Objective 4: Personalized Disease Advisory

Provide customized recommendations regarding pesticides, fertilizers, irrigation, nutrient management, and disease prevention.

Objective 5: Crop Health Monitoring

Maintain complete crop health history for every registered farmer and enable long-term monitoring.

Objective 6: Improve Agricultural Productivity

Increase crop yield through early disease diagnosis and timely treatment recommendations.

Objective 7: Reduce Pesticide Usage

Recommend scientifically approved pesticides only when necessary, thereby minimizing environmental pollution.

Objective 8: Support Sustainable Agriculture

Encourage eco-friendly farming practices by reducing excessive chemical usage and promoting integrated disease management.

Objective 9: Generate Agricultural Analytics

Produce dashboards and reports showing disease trends, regional outbreaks, prediction accuracy, and farmer activities.

Objective 10: Multilingual Accessibility

Provide support for multiple regional languages to improve accessibility among rural farming communities.

Objective 11: Continuous AI Improvement

Improve prediction accuracy continuously using newly collected datasets and expert feedback.

Objective 12: Government Decision Support

Provide analytical reports that help government agencies monitor disease outbreaks and formulate agricultural policies.

Table 1.1 Project Objectives

Objective ID	Objective	Expected Outcome
OBJ-1	Early Disease Detection	Reduce crop damage
OBJ-2	AI Disease Prediction	Improve prediction accuracy
OBJ-3	Mobile Accessibility	Easy access for farmers
OBJ-4	Personalized Advisory	Better farming decisions
OBJ-5	Crop Monitoring	Disease history management
OBJ-6	Yield Improvement	Increase agricultural productivity
OBJ-7	Reduce Pesticides	Lower production cost
OBJ-8	Sustainable Farming	Environmental protection
OBJ-9	Analytics	Better agricultural planning
OBJ-10	Multilingual Support	Improved accessibility
OBJ-11	AI Learning	Higher prediction accuracy
OBJ-12	Government Support	Better policy decisions

1.4 Project Scope

The scope of the AI-Based Crop Disease Detection and Farmer Advisory Platform covers the complete lifecycle of crop disease identification, advisory generation, historical record management, and agricultural analytics.

The software enables farmers to securely register and authenticate themselves before accessing the application. After successful login, farmers can upload images of crop leaves captured through mobile phones or digital cameras. The uploaded images are preprocessed using image enhancement techniques before being analyzed by trained deep learning models.

The AI engine identifies crop diseases by comparing extracted image features with thousands of labeled training samples. Once disease prediction is completed, the system displays the disease name, confidence percentage, disease description, probable causes, and severity level.

Based on the detected disease, the advisory module recommends appropriate pesticides, fertilizers, irrigation schedules, nutrient management techniques, and preventive farming practices. Farmers also receive suggestions regarding disease prevention, crop maintenance, and future monitoring activities.

The platform stores prediction history, enabling farmers to review previous diagnoses and compare crop health over different cultivation seasons. Agricultural experts can review prediction accuracy, update disease information, and improve advisory recommendations whenever required.

Administrators manage user accounts, disease databases, crop information, AI models, cloud storage, reports, notifications, and overall system security. The platform also generates analytical dashboards showing disease distribution, prediction statistics, crop performance, and regional disease outbreaks.

Future enhancements include weather forecasting integration, IoT-based soil monitoring, drone image analysis, satellite crop surveillance, blockchain-enabled agricultural records, AI chatbots, voice assistants, crop market prediction, and smart irrigation automation.

Table 1.2 Project Scope

Included Scope	Description
User Registration	Farmer account creation
Login System	Secure authentication
Image Upload	Upload crop leaf images

AI Prediction	Disease detection using AI
Advisory Module	Fertilizer and pesticide recommendation
History Module	Previous disease records
Expert Validation	Verify AI prediction
Reports	Disease analysis reports
Notifications	Disease alerts
Dashboard	Crop monitoring dashboard

ANNEXURE A – AGILE SPRINT PLANNING & USER STORY WORKSHOP

AI-BASED CROP DISEASE DETECTION AND FARMER ADVISORY PLATFORM

1.1 INTRODUCTION

Crop diseases are a major threat to agricultural productivity and food security. Early detection of diseases can significantly reduce crop loss, lower production cost and improve yield. However, farmers often lack timely access to experts for accurate diagnosis and treatment.

This project proposes an AI-Based Crop Disease Detection and Farmer Advisory Platform that enables farmers to upload crop leaf images and get instant disease prediction along with treatment, fertilizer, and preventive recommendations.

The system leverages Deep Learning, Image Processing and a knowledge base of agricultural experts to deliver accurate, reliable and multilingual advisory services. The software will be developed using Agile methodology with iterative sprints to continuously deliver value and adapt to changing requirements.

1.2 PROJECT VISION

To empower every farmer with intelligent, accessible and reliable technology that enables early detection of crop diseases and provides personalized advisory services for healthier crops, higher productivity and sustainable agriculture.

1.3 PROJECT OBJECTIVES

Objective ID	Objective Description	Expected Outcome
OBJ-1	Early disease detection using AI	Reduce crop loss
OBJ-2	Accurate disease classification	High prediction accuracy
OBJ-3	Easy to use mobile & web application	Accessibility for all farmers
OBJ-4	Personalized advisory (pesticide, fertilizer, irrigation)	Better farming decisions
OBJ-5	Maintain crop health history	Track and monitor crop health
OBJ-6	Improve agricultural productivity	Increase crop yield
OBJ-7	Reduce unnecessary pesticide usage	Lower cost & pollution
OBJ-8	Support sustainable farming	Eco-friendly agriculture
OBJ-9	Generate analytics and reports	Data-driven decisions
OBJ-10	Provide multilingual support	Reach farmers in native language
OBJ-11	Continuously improve AI model	Better accuracy over time
OBJ-12	Support government decision making	Better agricultural policies

1.4 PROJECT SCOPE

The system covers the end-to-end process from crop image upload to disease prediction and advisory. It also includes user management, expert validation, historical records, report generation and notifications.

INCLUDED IN SCOPE	DESCRIPTION
User Management	Farmer, Expert and Admin registration & authentication
Image Upload & Processing	Capture/upload crop images and preprocess them
AI Disease Prediction	Detect disease using deep learning models
Advisory Module	Provide pesticide, fertilizer, irrigation & preventive advice
History & Monitoring	Store and view previous predictions and crop history
Expert Validation	Experts verify and update disease information
Reports & Analytics	Generate disease statistics and analytical reports
Notifications	Send alerts for disease outbreaks and reminders

1.5 WORKFLOW DIAGRAM

1.6 OVERALL SYSTEM ARCHITECTURE

1.7 CROP TYPES & FEATURES

The system supports multiple crop types and provides disease detection for various categories.

1.8 BENEFITS TO STAKEHOLDERS

Stakeholder	Benefits
Farmers	Early disease detection, higher yield, lower cost, easy access.
Agricultural Experts	Platform to share knowledge and validate predictions
Government	Monitor disease outbreaks, make better policies
Researchers	Access to datasets for research and model improvement
Environment	Reduced chemical usage, eco-friendly agriculture
Society	Better food quality and food security

1.9 TECHNOLOGY STACK

Frontend : Flutter / React / HTML / CSS

Backend : Python (Django/Fast) / Node.js

AI/ML : TensorFlow / Keras / OpenCV

Database : MySQL / PostgreSQL

Cloud : AWS / Firebase

Tools : Git, Docker, Jenkins

1.11 SUCCESS METRICS

Prediction Accuracy
Target accuracy above 90%

User Satisfaction
Positive feedback from farmers

Adoption Rate
Increasing number of active users

Reduced Crop Loss
Decrease in disease-related losses

Timely Advisory
Recommendations provided in real-time

System Performance
High uptime and fast response time

Technology will not replace great farmers, but technology in the hands of great farmers can be transformational. – Unknown

STAKEHOLDER ANALYSIS AND PROJECT ORGANIZATION

1.5 Stakeholder Identification

Stakeholders are individuals, groups, or organizations who influence or are influenced by the successful development and implementation of the AI-Based Crop Disease Detection and Farmer Advisory Platform. Their active participation ensures that the software satisfies business requirements, user expectations, and technical objectives. Identifying stakeholders at the beginning of the project helps the Agile development team understand customer needs, prioritize requirements, and deliver maximum business value during each sprint.

The proposed system includes both primary and secondary stakeholders. Primary stakeholders directly interact with the application, whereas secondary stakeholders benefit indirectly from the system's successful implementation.

1.6 Primary Stakeholders

1. Farmer

Farmers are the primary users of the platform. They use the application to upload crop leaf images, detect crop diseases, receive advisory recommendations, and maintain crop health records.

Responsibilities

- Register and log into the application.
 - Upload crop images.
 - View AI disease prediction.
 - Follow treatment recommendations.
 - Monitor crop health history.
 - Provide feedback regarding system usability.
-

Expected Benefits

- Early disease detection.
 - Increased crop yield.
 - Reduced farming expenses.
 - Improved decision-making.
 - Easy access to expert guidance.
-

2. Agricultural Expert (Agronomist)

Agricultural experts ensure the scientific accuracy of disease diagnosis and treatment recommendations.

Responsibilities

- Verify AI-generated disease predictions.
 - Update disease database.
 - Recommend pesticides and fertilizers.
 - Review incorrect AI predictions.
 - Improve agricultural knowledge base.
-

Expected Benefits

- Faster farmer support.
 - Efficient knowledge sharing.
 - Improved disease monitoring.
 - Better advisory quality.
-

3. System Administrator

The administrator manages the complete software platform.

Responsibilities

- Create user accounts.
 - Manage databases.
 - Update disease information.
 - Monitor server performance.
 - Generate reports.
 - Manage security.
 - Perform system backup.
-

Expected Benefits

- Efficient platform management.

- Secure data storage.
 - Better system availability.
-

4. Product Owner

The Product Owner represents business requirements and customer expectations throughout Agile development.

Responsibilities

- Gather customer requirements.
 - Prioritize Product Backlog.
 - Define Sprint Goals.
 - Approve completed features.
 - Communicate with stakeholders.
-

Expected Benefits

- Maximum business value.
 - Better customer satisfaction.
 - Faster requirement implementation.
-

5. Scrum Master

The Scrum Master ensures Agile practices are followed throughout software development.

Responsibilities

- Conduct Sprint Planning.
 - Organize Daily Scrum.
 - Remove project obstacles.
 - Conduct Sprint Review.
 - Conduct Sprint Retrospective.
 - Improve team productivity.
-

Expected Benefits

- Better collaboration.

- Smooth sprint execution.
 - Higher software quality.
-

6. Development Team

The development team consists of software engineers responsible for designing, developing, testing, and deploying the application.

Team Members

- Frontend Developer
 - Backend Developer
 - Database Developer
 - AI Engineer
 - UI/UX Designer
 - QA Engineer
 - DevOps Engineer
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Responsibilities

- Software design.
 - Coding.
 - Database implementation.
 - AI model development.
 - Testing.
 - Deployment.
 - Maintenance.
-

1.7 Secondary Stakeholders

Government Agriculture Department

Government agencies utilize reports generated by the platform to monitor disease outbreaks, improve agricultural planning, and formulate farmer welfare policies.

Agricultural Universities

Universities use crop disease datasets for research and AI model improvement.

Research Organizations

Researchers improve disease prediction algorithms and develop new AI technologies.

Cloud Service Provider

Provides secure cloud infrastructure, storage, database hosting, and application deployment services.

Internet Service Provider

Ensures uninterrupted network connectivity between farmers and cloud servers.

Pesticide Manufacturers

Provide scientifically approved pesticides recommended by agricultural experts.

Fertilizer Companies

Provide crop-specific fertilizers recommended by the advisory engine.

Financial Institutions

Offer agricultural loans based on crop monitoring information.

Insurance Companies

Use disease reports for crop insurance claim verification.

1.8 Stakeholder Analysis Table

Stakeholder	Interest	Influence	Responsibility
Farmer	Very High	High	Use the application
Agricultural Expert	High	High	Verify disease prediction
Product Owner	Very High	High	Manage product backlog
Scrum Master	High	Medium	Manage Agile process
Development Team	High	High	Build the software
AI Engineer	High	High	Train AI models
Administrator	High	High	Manage system
Government	Medium	Medium	Monitor agriculture

Researchers	Medium	Low	Improve AI
Cloud Provider	Medium	Medium	Provide hosting

1.9 Stakeholder Power–Interest Matrix

Stakeholder	Power	Interest	Strategy
Product Owner	High	High	Manage Closely
Farmers	High	High	Manage Closely
Agricultural Experts	High	High	Manage Closely
Development Team	High	High	Manage Closely
Scrum Master	High	High	Manage Closely
Administrator	Medium	High	Keep Involved
Government Department	High	Medium	Keep Satisfied
Researchers	Medium	Medium	Keep Informed
Cloud Provider	Medium	Medium	Keep Informed
Public	Low	Low	Monitor

1.10 Stakeholder Communication Plan

Stakeholder	Communication Method	Frequency	Responsible Person
Farmers	Mobile App Notifications	Daily	Product Owner
Agricultural Experts	Meetings	Weekly	Scrum Master
Development Team	Daily Scrum	Daily	Scrum Master
Product Owner	Sprint Review	Every Sprint	Scrum Team
Government	Reports	Monthly	Administrator
Researchers	Email	Quarterly	AI Engineer

Cloud Provider	Technical Meetings	Monthly	DevOps Engineer
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1.11 RACI Matrix

Activity	Product Owner	Scrum Master	Developer	AI Engineer	Admin	Farmer
Requirement Gathering	A	C	I	I	I	R
Sprint Planning	A	R	C	C	I	I
UI Development	I	C	R	I	I	I
AI Model Training	I	C	C	R	I	I
Database Design	I	C	R	I	I	I
Testing	C	C	R	R	I	I
Deployment	I	C	R	C	R	I
Maintenance	I	I	C	C	R	I

Legend:

- **R** – Responsible
- **A** – Accountable
- **C** – Consulted
- **I** – Informed

1.12 Stakeholder Engagement Strategy

The Agile team follows a collaborative engagement strategy to ensure continuous stakeholder participation throughout the project. Farmers are involved during requirement gathering, usability testing, Sprint Reviews, and product demonstrations. Agricultural experts validate AI predictions and improve advisory recommendations based on scientific knowledge. The Product Owner prioritizes requirements based on stakeholder feedback, while the Scrum Master facilitates Agile ceremonies and resolves project impediments.

Regular Sprint Reviews provide stakeholders with opportunities to evaluate completed features and suggest improvements. Daily Scrum meetings ensure effective communication among development team members. Sprint Retrospectives help identify areas for continuous process improvement, enabling the team to enhance productivity and software quality in subsequent sprints.

CREATE EPICS AND USER STORIES

2.1 Introduction

In Agile Software Development, an **Epic** is a large body of work that represents a major business functionality. Epics are divided into smaller and manageable **User Stories**, which describe software features from the perspective of end users. User Stories help the Scrum Team understand customer requirements, estimate development effort, prioritize work, and deliver software incrementally through sprints.

For the **AI-Based Crop Disease Detection and Farmer Advisory Platform**, the project requirements are divided into several Epics such as User Management, Crop Image Upload, Disease Detection, Farmer Advisory, Notifications, Reporting, and Administration. Each Epic contains multiple User Stories following the Agile format:

As a <user>, I want <feature>, so that <benefit>.

These User Stories form the Product Backlog and guide Sprint Planning.

2.2 Epic 1 – User Registration and Authentication

Epic Description

This Epic focuses on allowing farmers, agricultural experts, and administrators to securely register, log in, and manage their user accounts.

Business Value

- Secure access to the platform.
 - Personalized user experience.
 - Protection of user information.
-

User Story 1

As a farmer, I want to register my account, so that I can access crop disease detection services.

Priority: High

User Story 2

As a farmer, I want to log in securely, so that my crop information remains protected.

Priority: High

User Story 3

As a farmer, I want to reset my password, so that I can recover my account if I forget my password.

Priority: Medium

User Story 4

As a farmer, I want to edit my personal profile, so that my farming information remains updated.

Priority: Medium

User Story 5

As an administrator, I want to manage user accounts, so that unauthorized users cannot access the system.

Priority: High

Table 2.1 Epic 1 Summary

User Story ID	User Story	Priority
US-01	Farmer Registration	High
US-02	Secure Login	High

US-03	Password Recovery	Medium
US-04	Profile Update	Medium
US-05	User Management	High

2.3 Epic 2 – Crop Image Upload

Epic Description

The platform allows farmers to upload crop leaf images for AI analysis. High-quality image upload is essential for accurate disease detection.

Business Value

- Enables AI disease prediction.
 - Simplifies farmer interaction.
 - Improves prediction accuracy.
-

User Story 6

As a farmer, I want to upload crop leaf images, so that AI can detect crop diseases automatically.

Priority: High

User Story 7

As a farmer, I want to capture images using my mobile camera, so that I can upload images instantly from the field.

Priority: High

User Story 8

As a farmer, I want to preview uploaded images, so that I can verify image quality before submission.

Priority: Medium

User Story 9

As a farmer, I want to delete incorrect images, so that only accurate images are analyzed.

Priority: Medium

User Story 10

As the system, I want to validate uploaded images, so that poor-quality images are rejected automatically.

Priority: High

Table 2.2 Epic 2 Summary

User Story ID	User Story	Priority
US-06	Upload Crop Image	High
US-07	Camera Capture	High
US-08	Preview Image	Medium
US-09	Delete Image	Medium
US-10	Image Validation	High

2.4 Epic 3 – AI-Based Disease Detection

Epic Description

This Epic enables the Artificial Intelligence engine to analyze uploaded images and accurately identify crop diseases.

Table 2.4 Epic 4 Summary

User Story ID	User Story	Priority
US-16	Pesticide Recommendation	High
US-17	Fertilizer Recommendation	High

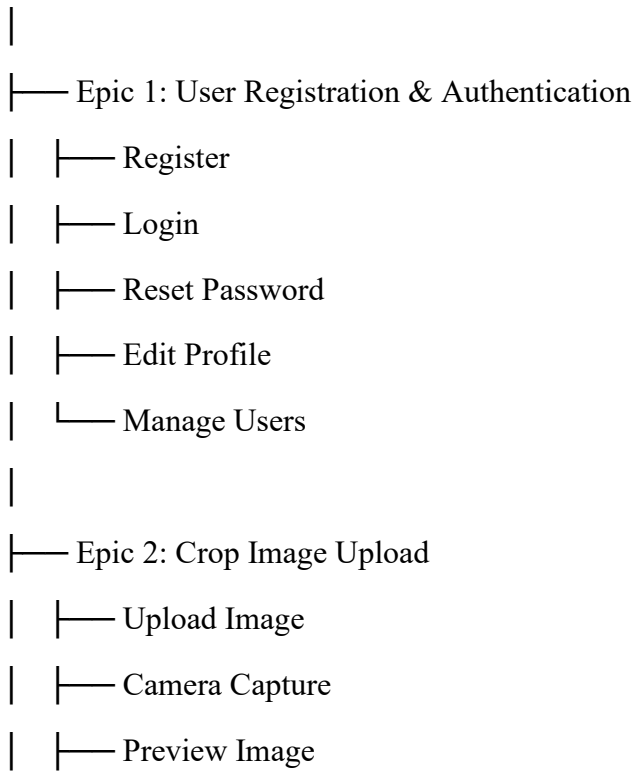
US-18	Irrigation Advice	Medium
US-19	Disease Prevention	Medium
US-20	Expert Advisory	High

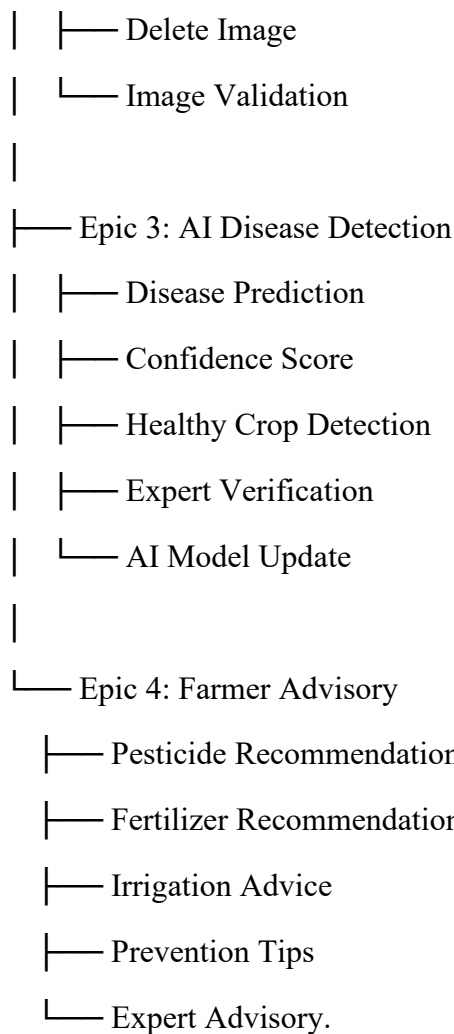
2.6 Summary of User Stories (1–20)

Epic	Number of User Stories
Epic 1 – User Registration & Authentication	5
Epic 2 – Crop Image Upload	5
Epic 3 – AI Disease Detection	5
Epic 4 – Farmer Advisory	5
Total	20 User Stories

Figure 2.1 – Epic Breakdown (Insert in Report)

AI-Based Crop Disease Detection Platform





PART 2B – Remaining Epics & User Stories (21–40)

Epic 5 – Notifications

- **US-21:** As a farmer, I want disease alerts, so that I can take preventive action.
 - **US-22:** As a farmer, I want weather notifications, so that I can protect my crops.
 - **US-23:** As a farmer, I want irrigation reminders, so that I can water crops on time.
 - **US-24:** As an admin, I want to send announcements, so that all users receive important updates.
 - **US-25:** As a farmer, I want fertilizer reminders, so that I do not miss applications.
-

Epic 6 – Reports & Dashboard

- **US-26:** As a farmer, I want to view disease history, so that I can monitor crop health.
- **US-27:** As an expert, I want analytics reports, so that I can study disease trends.
- **US-28:** As an admin, I want a dashboard, so that I can monitor platform usage.

- **US-29:** As an admin, I want prediction statistics, so that I can evaluate AI performance.
- **US-30:** As a farmer, I want downloadable reports, so that I can keep records.

Epic 7 – Administration

- **US-31:** As an admin, I want to manage crops.
- **US-32:** As an admin, I want to manage diseases.
- **US-33:** As an admin, I want to manage pesticides.
- **US-34:** As an admin, I want to manage fertilizers.
- **US-35:** As an admin, I want to manage user feedback.

Epic 8 – Security & Maintenance

- **US-36:** As a user, I want secure login.
 - **US-37:** As an admin, I want encrypted data storage.
 - **US-38:** As an admin, I want automatic backup.
 - **US-39:** As an admin, I want audit logs.
 - **US-40:** As a developer, I want system monitoring.
-

3. Acceptance Criteria (Sample)

User Story	Acceptance Criteria (Given–When–Then)
US-01 Registration	Given a new user, When valid details are entered, Then registration succeeds.
US-06 Upload Image	Given a logged-in farmer, When an image is uploaded, Then the system stores it successfully.
US-11 Disease Detection	Given a valid image, When AI processes it, Then disease name and confidence are displayed.
US-16 Advisory	Given a detected disease, When prediction is complete, Then recommendations are displayed.
US-26 Disease History	Given previous records exist, When the farmer opens history, Then all previous detections are shown.

4. Product Backlog

Priority	Backlog Item
High	User Registration
High	Login
High	Image Upload
High	AI Disease Detection
High	Advisory System
Medium	Notifications
Medium	Reports
Medium	Dashboard
Low	Feedback Module
Low	Settings

5. Sprint Backlog (Sprint 1)

User Story	Development Tasks
Registration	UI, Database, Validation
Login	Authentication, Session Management
Upload Image	Camera Integration, File Upload
Disease Detection	AI API Integration
Advisory	Recommendation Module

6. Story Point Estimation (Fibonacci)

User Story	Story Points	Justification
Registration	3	Simple form
Login	2	Authentication

Upload Image	5	Camera + Upload
Disease Detection	8	AI Integration
Advisory	5	Business Logic
Notifications	3	Push Notification
Reports	5	Report Generation
Dashboard	5	Data Visualization

Estimation Technique: Fibonacci Sequence (1, 2, 3, 5, 8, 13)

Justification:

-
- **2–3 Points:** Low complexity
 - **5 Points:** Medium complexity
 - **8 Points:** High complexity requiring AI integration
-

7. Sprint Goal

Sprint

Goal:

Develop a Minimum Viable Product (MVP) that enables farmers to register, log in, upload crop images, receive AI-based disease detection, and view advisory recommendations.

Expected Sprint Deliverables

-
- Farmer Registration Module
 - Login Module
 - Crop Image Upload Module
 - AI Disease Detection Module
 - Farmer Advisory Module
 - Basic Dashboard
 - Database Integration
 - Admin Login
 - Sprint Review Report

- Sprint Retrospective Report
-

Definition of Ready (DoR)

A user story is ready when:

-
- Requirements are clearly defined.
 - Acceptance criteria are available.
 - Dependencies are identified.
 - Story points are estimated.
 - Product Owner approves the story.
-

Definition of Done (DoD)

A user story is done when:

-
- Coding is complete.
 - Unit testing passes.
 - Integration testing passes.
 - No critical bugs remain.
 - Product Owner approves the feature.
 - Documentation is updated.
-

Sprint Review

The completed features are demonstrated to stakeholders. Feedback is collected and added to the Product Backlog for future sprints.

Sprint Retrospective

The Scrum Team discusses:

-
- What went well.
 - What challenges were faced.
 - What improvements should be made in the next sprint.
-

Conclusion

The **AI-Based Crop Disease Detection and Farmer Advisory Platform** applies Agile principles to deliver software incrementally. By organizing requirements into Epics, User Stories, Product Backlog, Sprint Backlog, and Story Points, the project ensures continuous stakeholder involvement, rapid delivery, and high-quality software. The platform supports farmers with AI-driven disease detection, timely advisory services, and sustainable agricultural practices, ultimately improving crop productivity and reducing farming losses.



Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.

Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.
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Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25